



Sex Differences in Concurrent Aerobic and Strength Training

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Introduction

As illustrated by the various chapters throughout this book, there are many different aspects to concurrent endurance and strength training. However, the most basic question is whether performing these two types of training concurrently will lead to different physiological adaptations compared to performing only one type of training. Most attention in the literature has been given to the potential negative effects on the traditional adaptations to a strength training program from concurrently performing endurance training, often called the interference effect. Furthermore, attention has also been given to the potential positive effects which endurance athletes may gain by adding strength training to their normal training. The latter will be discussed in depth in Part V.

Many studies have reported that concurrent training can lead to impaired neuromuscular adaptations as compared to strength training only [1–11]. Furthermore, adding strength training to endurance athletes' normal training has been reported to improve performance in many endurance sports [12–26]. However, the data are equivocal and many studies report no negative effect on strength-related adaptations after concurrent training [27–34], or no beneficial effects on endurance performance when endurance athletes additionally perform strength training [35–45]. The reasons for the different findings in these two aspects of concurrent training are discussed extensively in other parts of this book. The main focus of this chapter will be to discuss possible sex differences in the adaptations to concurrent training. Specifically this chapter aims to review whether the adaptations to concurrent

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training are sex-specific. Furthermore, it will be discussed if any sex differences are apparent in the effects of adding strength training to normal endurance training for endurance athletes.

Sex Differences in Performance and Relevant Physiology

Men are on average about 5–10% taller and 20–25% heavier than their female counterparts [46, 47]. In addition, men have about 40–50% and 30% more muscle mass in the upper and lower body, respectively, while women have approximately 20–25% of body fat compared to 13–16% in men [47, 48]. Furthermore, men have been reported to have larger mean muscle fiber areas than women both in the upper (m. biceps brachii) and lower body (m. vastus lateralis) [49].

As a result, review papers report that men are 40–55% stronger than women and this difference is especially pronounced in the upper body. Men also generate greater muscular power as well as higher anaerobic power compared to women [48]. The difference in absolute strength and anaerobic performance narrows when it is related to body mass and lean body mass [47], but it does not disappear.

Women have thinner ventricular walls with less myocardial tissue, smaller cavity sizes, and smaller stroke volume than men [48]. Furthermore, women have on average 10–16% lower hemoglobin concentrations [48] and a lower total hemoglobin mass both in absolute and relative values [50, 51] compared to men. Therefore, women have on average 15–30% lower maximal oxygen consumption ($\text{VO}_{2\text{max}}$) than men [48].

When performing muscle contractions at the same relative intensities, women have been shown to be more resistant to muscle fatigue than men [48, 52]. Since one of the proposed mechanisms for the interference effect is residual fatigue induced by the endurance training compromising the subsequent strength training [53, 54], it might be speculated that sex differences in fatigability will lead to sex differences in the response to concurrent training. The reason for the lower fatigability in women is not completely understood, but it is probably related to the fact that women exert less absolute force at a relative workload, leading to a smaller compression of the microcirculation and lower muscle oxygen demands [47, 52]. When lifting the same absolute load, muscular endurance is greater in men [47, 48].

Also, the substrate utilization during exercise differs between men and women [55, 56]. During endurance exercise at submaximal intensities, women oxidize proportionally more fat and less carbohydrates and proteins than men [55]. Since it has been suggested that low glycogen stores after endurance training can lead to less effective strength training sessions and impaired anabolic signaling after strength training [53, 54, 57], sex differences in substrate utilization during endurance training sessions can in theory lead to sex differences in the adaptations to strength training sessions performed concurrently.

There are also some hormonal differences between men and women, the most obvious being different secretion of sex hormones. The abovementioned differences in muscle mass and strength are usually explained by the higher serum testosterone

levels in men [48, 58], while the higher fat oxidation in woman has been attributed to estradiol secreted by the ovaries [55]. Testosterone levels in the blood may increase in both men and women in response to exercise, even though the increase often is smaller in women [59]. Exercise may also increase estradiol levels in women [59].

The menstrual cycle has a profound effect on the secretion of the ovarian hormones in women [60]. During the follicular phase there is a gradual increase in estrogen concentrations, while progesterone dominates during the luteal phase. However, it seems that the different phases of the menstrual cycle affect maximal strength, muscle fatigability, and maximal aerobic power to a small degree only, even though conflicting results exist [60, 61]. The same seems to be true for most physiological responses to submaximal exercise, even though increased oxygen consumption (VO_2) at high intensities during the mid-luteal phase has been reported [60].

Based on the differences in physical performance and hormone levels between men and women, it might be speculated that there are sex differences in the adaptations to strength and endurance training when they are performed separately. However, the typical muscular adaptations to strength training seems to be similar between the sexes, as studies show that there are no differences in the relative increase in maximal strength [62, 63], increases in muscle mass [63], and the fiber type shift from type IIX towards type IIA after strength training [62]. The same is true for the relative increase in $\text{VO}_{2\text{max}}$ after endurance training [64].

Sex Differences in Concurrent Training

Unfortunately, most of the research dealing with concurrent training has been carried out using male participants. In addition, there is a lack of studies that directly compare the effect of concurrent training between men and women. Therefore, this chapter will mainly be based on results from separate studies performed in male and female participants or results from the few studies that include both female and male participants. The challenge with the first approach is that different studies using male and female participants usually also differ in many other methodological aspects that might be relevant. Similarly, the challenge with the second approach is that most of these studies include both men and women to get a larger number of participants and the total number of each sex are often too small for meaningful comparisons. Therefore, these studies usually report pooled data from the male and female subgroups.

Are There Any Sex Differences in the Effects of Concurrent Training on the Adaptations to Strength Training?

Above it was shortly discussed that many studies report attenuated adaptations in various aspects of muscle strength and muscle growth after concurrent training, even though the results are conflicting. There are numerous variables that are

possible to manipulate in the design of these studies and the conflicting results are probably related to many different methodological aspects that vary between studies. For example, it seems that the improvement in maximal power and explosive strength are more attenuated than maximal strength and muscle hypertrophy after concurrent training [11, 65, 66]. It might also be speculated that the use of male and female participants might explain some of the different findings. Unfortunately, there are not many studies in this area that are performed with female participants [29, 67–69], and some of these studies do not directly investigate if an interference effect exists [68, 69]. However, the studies investigating the interference effect using female participants do not report compromised neuromuscular adaptations [29, 67]. Thus, based on these studies it might be concluded that there is no interference effect or that it at least is smaller in women compared to men. However, it should be noted that in these studies the total training volume was quite low and no impairment in training adaptations is also common in male participants performing a low total training volume or a low number of training sessions per week [27, 28, 30–34]. Therefore, it is difficult to interpret if the lack of an interference effect is sex-specific or just simply due to a low training volume or other aspects related to the study design.

In recent work from our group we investigated if well-trained female endurance athletes with no strength training experience adding 11 weeks of strength training to their training would have different changes in muscle cross sectional area (CSA), maximal strength, and explosive strength compared to a group of untrained women performing the same heavy strength training program [70]. We observed similar increases in the lean mass of the legs, one repetition maximum (RM) in one-legged leg press, and maximal isometric knee extension torque in both groups. However, the improvement in isokinetic torque at a contraction velocity of $240^{\circ} \text{ s}^{-1}$ and the improvement in squat jump height were smaller in the female endurance athletes compared to the untrained participants. The results from this study can be compared to a study including male participants with a similar design and strength training program [71]. Here it was reported that well-trained male cyclists showed smaller increases in 1RM and CSA of the thigh muscles, squat jump height, and maximal rate of force development (RFD) after 12 weeks of concurrent training compared to a group of recreational active individuals only performing strength training. These two studies again suggest that the magnitude of the interference effect may be smaller in women than in men, at least in terms of maximal strength at slow contraction velocities and muscle growth. However, the different findings in these two studies might again be explained by the amount of endurance training performed. The female athletes performed about 5 h of endurance training per week, whereas the male cyclists performed endurance training for about 10 h per week. It is therefore unclear if the different findings are explained by the amount of endurance training or solely by sex. Reports that 1–3 h of weekly endurance training in recreationally active male and female runners did not lead to any impaired strength adaptations [5] suggest that the amount of endurance training probably is most important. Furthermore, the smaller improvement in power-related measurements in the female athletes compared to the untrained individuals illustrates that an interference effect

also occurred in the female athletes [70]. The finding that negative effects of concurrent training are most pronounced in the development of power and explosive strength compared to maximal strength is in accordance with studies using predominantly male participants [65, 66].

As described, there are studies that investigate the concurrent training phenomenon that include small subsets of both male and female participants [1, 2, 4, 5, 8, 27, 65, 72]. Unfortunately, most of these studies report pooled data for both sexes. The reason for this is usually that preliminary analyses showed that there were similar patterns of results for the male and female participants [1, 5, 8, 65, 72]. Furthermore, other of these studies report no sex differences in regard to the interference effect [2]. Even though these analyses are carried out with small sample sizes, these studies further indicate that the interference effect is similar between the sexes.

Based on the considerations above it appears that the possible interference effect of concurrent strength and endurance in most cases are similar between men and women. However, there might be other aspects of concurrent training that can differ, for example how to optimally organize concurrent training for the smallest possible interference effect to occur. Two recent studies, including both male and female participants, investigated if it is best to perform strength and endurance training on different days or on the same day, and whether the exercise sequence matters when aerobic and strength training are performed in the same session [73, 74]. These studies reported similar changes in both men and women after the different training modes and no sex differences in which of the training modes that were best for the measured outcomes. Further support for this is given when comparing separate studies with male and female participants using almost identical training programs [68, 75–77]. Therefore, it seems that the optimal organization of concurrent training is similar between men and women.

Are There Any Sex Differences in the Effects of Strength Training on Performance in Endurance Sports?

As described above, adding strength training to a normal endurance training routine has been reported to improve running performance in some studies, while other studies did not report any beneficial effects. However, there do not appear to be any sex differences in the effects of strength training on running performance based on these studies. Improved running performance or running economy has been reported in studies using male [13, 15, 18, 19, 78] and female athletes [37, 79], as well as in studies reporting pooled data from both male and female athletes [12, 16, 80, 81]. Furthermore, no effect of strength training on running performance has also been reported in male [36, 39] and female athletes [82] or mixed groups of athletes [35, 83]. Even though the abovementioned studies do not indicate any sex differences in the effects of strength training on running performance, none of these studies were designed to directly answer this question. One of the few studies including a large enough number of both male and female runners to make meaningful comparisons is the study by Barnes et al. [12]. In this study, the effect of 7–10 weeks of

traditional heavy strength training or heavy strength training in combination with plyometric training on laboratory measures of performance and performance during the competition season in male and female collegiate runners was assessed. This study reported that both the male and female runners had improvements in peak treadmill running speed and running economy, and that both sexes had larger improvement in these measures after traditional heavy strength training. However, it was only the female runners that improved actual competitions times and this was evident after both types of strength training. Both types of strength training were possibly harmful to competition times in men. The apparent sex difference in this study is somewhat difficult to explain but the authors speculate that it might be related to a larger proportion of training above 80% of $\text{VO}_{2\text{max}}$ and shorter race distances in the female runners. However, the results may also indicate that the effect of strength training on real-world running performance is somewhat larger in women than in men.

Improved performance after addition of strength training to normal endurance training has also been reported in cycling, but as described, the results are equivocal. Unfortunately, to the best of the author's knowledge, there are no studies comparing the effects of adding strength training to endurance athlete's normal training on cycling performance between men and women. In addition, until recently, almost all research has been performed using male cyclists, and one of the few studies including female cyclists reported no improvement in cycling performance or cycling economy [41]. However, the cyclists in this study only included one strength training exercise twice a week making the training volume quite low. This might explain the lack of improved performance since studies using low strength training volume usually do not report any enhanced cycling performance even in men [44, 84].

Therefore, some years ago, our group conducted a more thorough investigation on the effects of strength training on various aspects of cycling performance and performance determinants in female cyclists [25, 26]. Furthermore, we used a strength training program identical to what has previously been reported to improve performance in male cyclists [21, 23]. The strength training program lasted for 12 weeks with two sessions per week with a training load of 10–4 RM. The effects of 12 weeks of heavy strength training on muscle strength, muscle hypertrophy, and various aspects of cycling performance were very similar in both the male [21, 23] and the female cyclists [25, 26] in these studies, and the results from these studies are summed up in Table 20.1. In neither men nor women did strength training have any effect on $\text{VO}_{2\text{max}}$ compared to control cyclists only performing their normal endurance training. Furthermore, performance in a 40 min all-out test, lactate threshold, and performance in the Wingate test improved similarly in both male and female cyclists after strength training. On the other hand, strength training did only improve cycling economy in the female cyclists. However, this is likely because the male cyclists were on a somewhat higher performance level than the female cyclists and it appears that it is difficult to improve cycling economy in very well trained cyclists [20]. There have been reported improved cycling economy after heavy strength training in male cyclists on a lower performance level [17]. To simulate a typical mass start race in cycling we performed a test consisting of a 3 h submaximal cycling trial

Table 20.1 Percent change in strength and cycling performance after 12 weeks of heavy strength training in male and female cyclists

	Male cyclists	Female cyclists
IRM	26.0 ± 6.6%	38.6 ± 19.0%
CSA	4.6 ± 1.7%	7.4 ± 5.3%
VO _{2max}	3.3 ± 4.6%	−1.1 ± 2.6% (ns)
Cycling economy	No change	3.5 ± 3.1%
Lactate threshold	4.1 ± 5.1%	7.3 ± 12%
40 min all-out	6.0 ± 5.6%	6.4 ± 7.9%
VO ₂ last hour, 3 h submax	−2.2 ± 2.0%	−3.3 ± 4.4%
5 min all-out	7.2 ± 6.6%	7.0 ± 4.5%
Wingate peak power	9.4 ± 9.6%	12.7 ± 12.6%
Wingate average power	1.2 ± 3.6% (ns)	3.4 ± 4.3%

The results from the male cyclists are from Ronnestad et al. [21, 23], and the results from the female cyclists are from Vikmoen et al. [25, 26]

Values are mean ± SD. *RM* repetition maximum, *VO_{2max}* maximal oxygen consumption, *VO₂ last hour, 3 h submax* oxygen consumption during the last hour of a 3 h submaximal cycling trial, *ns* not statistically significant change

followed by a 5 min all-out test. The effect of strength training on the results from this test was also strikingly similar between the male and female cyclists. Both had showed a lower VO₂ and hence an improved work economy during the last hour of the submaximal trial, compared to before the strength training intervention. This probably led to a lower magnitude of fatigue during the 3 h submaximal trial and both the male and female cyclists improved similarly in the 5 min all-out test after the strength training program (Table 20.1). None of these changes were observed in neither male nor female control cyclists. The results from these studies in both male and female cyclists indicate that there are no sex differences in the improvements in cycling performance when cyclists add strength training to their normal training. However, caution is warranted since these studies did not directly investigate if sex differences exist. Future studies should include sufficient numbers of both male and female cyclists that in parallel carries out the same strength training program to directly compare the effects between sexes.

Several adaptations to strength training have been suggested as important mechanisms for the improved cycling performance after strength training. These include an increased muscle CSA, leading to increased muscle strength, a muscle fiber transformation from type IIX towards type IIA, increased RFD that makes cyclists capable of reaching their peak torque earlier in the pedal stroke, reduced motor unit recruitment on the same absolute power output, and other neuromuscular factors [25, 26, 84, 85]. These adaptations to strength training appear to be similar between men and women and further support that the improvements in endurance performance resulting from these adaptations should be similar. For example, a fiber type shift from type IIX to type IIA has been reported in both male and female cyclists after heavy strength training [20, 25], and increases in muscle mass after strength training is also similar in men and women [63].

Summary

This chapter has discussed if the physical and physiological differences between men and women will lead to any sex differences in concurrent endurance and strength training. Men have more muscle mass and are leaner than age matched women. Furthermore, men are stronger, have greater muscular power, higher anaerobic power, and higher $\text{VO}_{2\text{max}}$ compared to women. However, women are less fatigable than men on the same relative intensities and have a better ability to burn fat on submaximal intensities. In addition, there are also hormonal differences between men and women.

Unfortunately, the existing literature is not sufficient to provide a definitive answer if there are sex differences in concurrent training. However, based on studies including participants of only one sex and studies including small numbers of participants of each sex indicate that the interference effects will be similar in both men and women, even though some findings points towards a smaller interference effect in women. It is more likely that other factors concerning the training program will be more important than sex to determine if an interference effect will occur. This is thoroughly discussed in other chapters of this book. Based on the current literature, it also appears that both male and female endurance athletes will similarly benefit from strength training added to their endurance training routine, even though one study indicates that female runners might have a better transformation of gains made in laboratory tests to real competition running performance. However, the research on sex differences in these areas is limited and more studies that directly investigate possible sex differences in concurrent training should be performed. These studies should include enough participants of both sexes that perform exactly the same concurrent training programs, so that meaningful comparisons between men and women can be made.

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